

## FULL OUTER JOIN

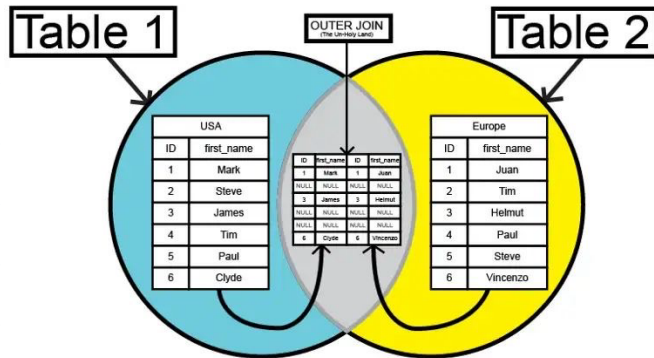
- returns:

↳ all matching rows

↳ all the rows that are matching on the common columns are combined into a single row.

↳ all unmatched rows

↳ where there's no match, we will put a NULL



Syntax:

```

SELECT col 1,
       col 2,
       col 3
FROM   table 1 AS A
FULL OUTER JOIN table 2 AS B
ON     A.common_column = B.common_column;
    
```

customers

| customer ID | customer Name |
|-------------|---------------|
| 1           | Ihabo         |
| 2           | Sarah         |
| 3           | Lerato        |
| 4           | John          |

orders

| order ID | customer ID | Product    |
|----------|-------------|------------|
| 101      | 1           | Laptop     |
| 102      | 2           | Phone      |
| 103      | 2           | Headphones |
| 104      | 5           | Tablet     |

```

SELECT customer ID,
       customer Name,
       order ID,
       product
FROM   customers AS c
FULL OUTER JOIN orders AS o
ON     c.customer ID = o.customer ID;
    
```

| customer ID | customer Name | order ID | Product    |
|-------------|---------------|----------|------------|
| 1           | Ihabo         | 101      | Laptop     |
| 2           | Sarah         | 102      | Phone      |
| 2           | Sarah         | 103      | Headphones |
| 3           | Lerato        | NULL     | NULL       |
| 4           | John          | NULL     | NULL       |
| 5           | NULL          | 104      | Tablet     |

removes duplicates      does not remove duplicates

UNION and UNION ALL

↳ combine the results of two SELECT statements by stacking the rows.

- The select statements must have the same number of columns.
- The columns must be in the same order.
- The columns must have the same data types.

current\_students

| student_id | StudentName |
|------------|-------------|
| 1          | Ihabo       |
| 2          | Sarah       |
| 3          | Lerato      |

new\_students

| StudentID | Student Name |
|-----------|--------------|
| 3         | Lerato       |
| 4         | John         |
| 5         | Mary         |

SELECT student\_id,  
StudentName,  
current\_students  
FROM  
UNION

SELECT student\_id,  
StudentName,  
new\_students  
FROM

| Student_id | StudentName |
|------------|-------------|
| 1          | Ihabo       |
| 2          | Sarah       |
| 3          | Lerato      |
| 4          | John        |
| 5          | Mary        |

current\_students

| student_id | StudentName |
|------------|-------------|
| 1          | Ihabo       |
| 2          | Sarah       |
| 3          | Lerato      |

new\_students

| StudentID | Student Name |
|-----------|--------------|
| 3         | Lerato       |
| 4         | John         |
| 5         | Mary         |

SELECT student\_id,  
StudentName,  
current\_students  
FROM  
UNION ALL

SELECT student\_id,  
StudentName,  
new\_students  
FROM

| Student_id | StudentName |
|------------|-------------|
| 1          | Ihabo       |
| 2          | Sarah       |
| 3          | Lerato      |
| 4          | John        |
| 5          | Mary        |
| 3          | Lerato      |